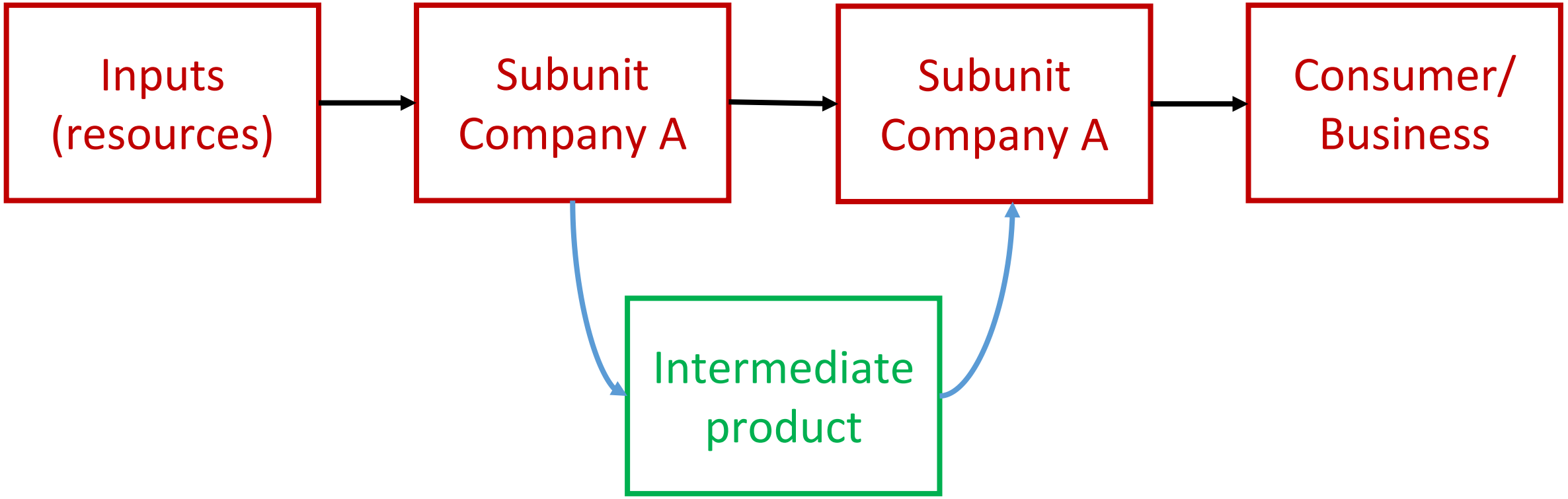
Topic 4: Transfer pricing

Sessions 6-7-8

Fundamentals

Transfer pricing



https://sabi-bvdinfo-com.ie.idm.oclc.org/Search.QuickSearch.serv?_CID

Search

sabi 2.600.000 Spanish and 800.000 Portuguese companies

Companies | Contacts | Industry research | News

Enter a company name, NIF or BvD ID

Alerts | Setting

Home > List (Standard list)

+ Show search strategy

1 of 3 | Note | PG | Columns | Save | Delete | Alerts | Export | Send

Companies with edited data are displayed in blue Modify

	Company Name		Country	Consolida Code	Last available year	Operating revenue / turnover th EUR Last avail. yr
1.	AMAZON SPAIN FULFILLMENT SLU	JANDO DE HENARES	SPAIN	U2	31/12/2019	333.289
2.	AMAZON ROAD TRANSPORT SPAIN SLU		SPAIN	U1	31/12/2019	237.682
3.	AMAZON ONLINE SPAIN SL.		SPAIN	U1	31/12/2019	159.664
4.	AMAZON SPAIN SERVICES SL.		SPAIN	U2	31/12/2019	106.626
5.	AMAZON DATA SERVICES SPAIN SL.		SPAIN	U1	31/12/2019	5.830

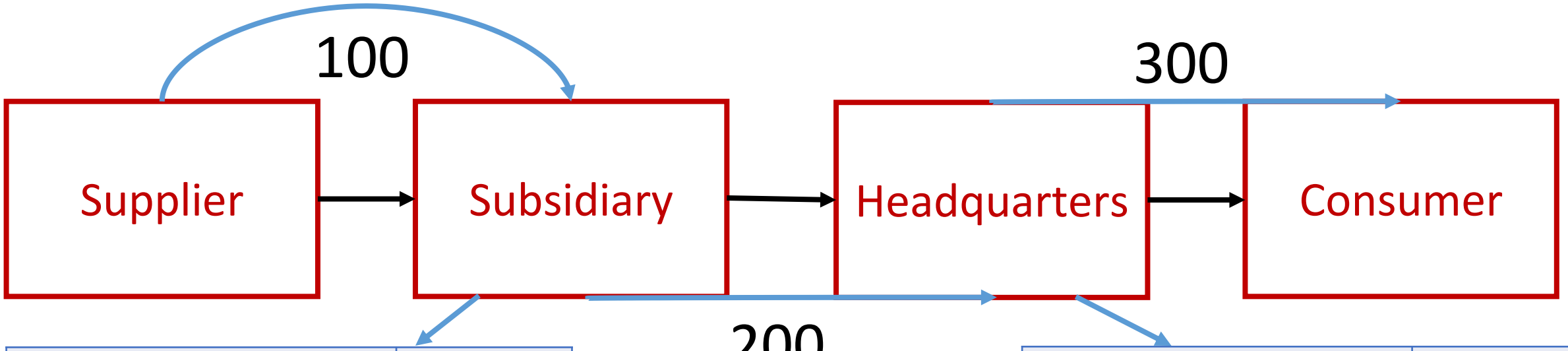
Transfer pricing -> PROFIT centers only

- Transfer pricing is **the price** one subunit of a company charges for the products/services it provides another subunit of the **same** company
- The transfer price creates revenues for the selling subunit and purchase cost to the buying subunit affecting their **operating income**
- **Intermediate product**: product or service transferred between subunits
- "Transfer Prices are the prices at which an enterprise transfers physical goods and intangible property or provides services to **associated** enterprises" (OECD Transfer price guidelines for multinational enterprises and tax administrations. July 2017).

Transfer prices should meet 4 criteria

- Promote **goal congruence** between subunits
- Induce managers to exert a high level of **effort**
- Help evaluate subunits' **performance**
- Preserves subunit **autonomy** (decentralized)

Why are transfer prices important?... Multinational 1

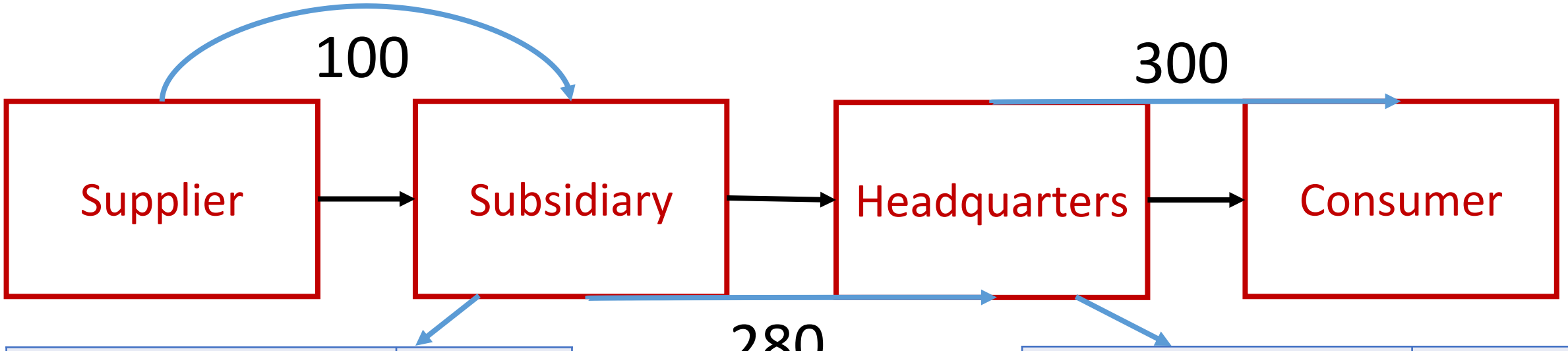


Revenues	200
- Costs	-100
Operating profits	100
- Tax 20%	-20
Net income	**80**

Overall income
120

Revenues	300
- Costs	-200
Operating profits	100
- Tax 60%	-60
Net income	**40**

Multinational 2

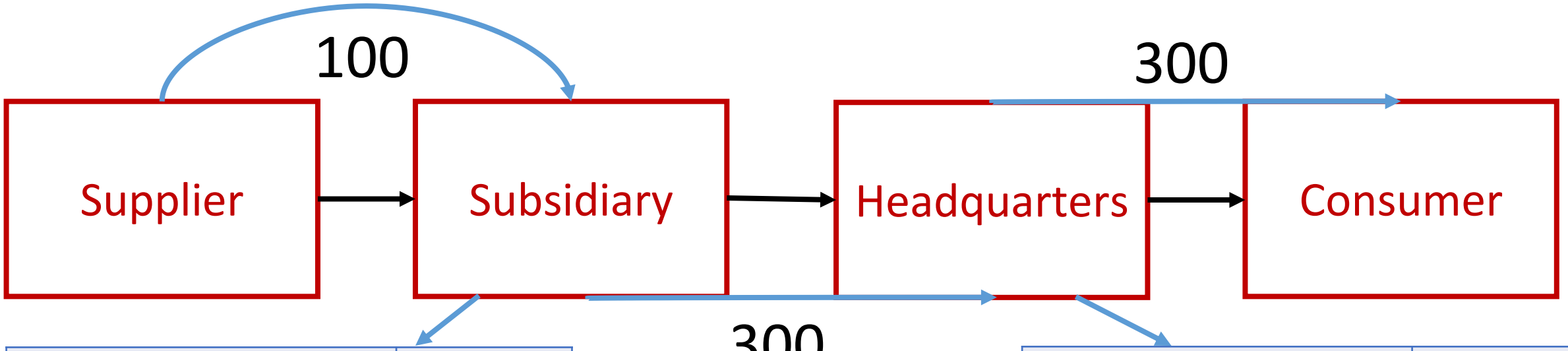


Revenues	280
- Costs	-100
Operating profits	180
- Tax 20%	-36
Net income	**144**

Overall income
152

Revenues	300
- Costs	-280
Operating profits	20
- Tax 60%	-12
Net income	**8**

Multinational 3

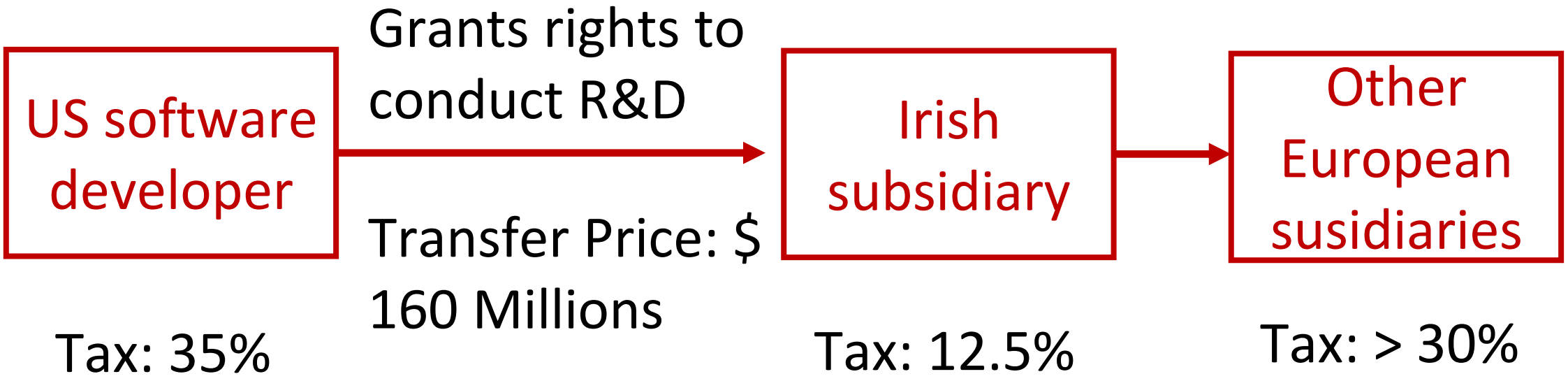


Revenues	300
- Costs	-100
Operating profits	200
- Tax 20%	-40
Net income	**160**

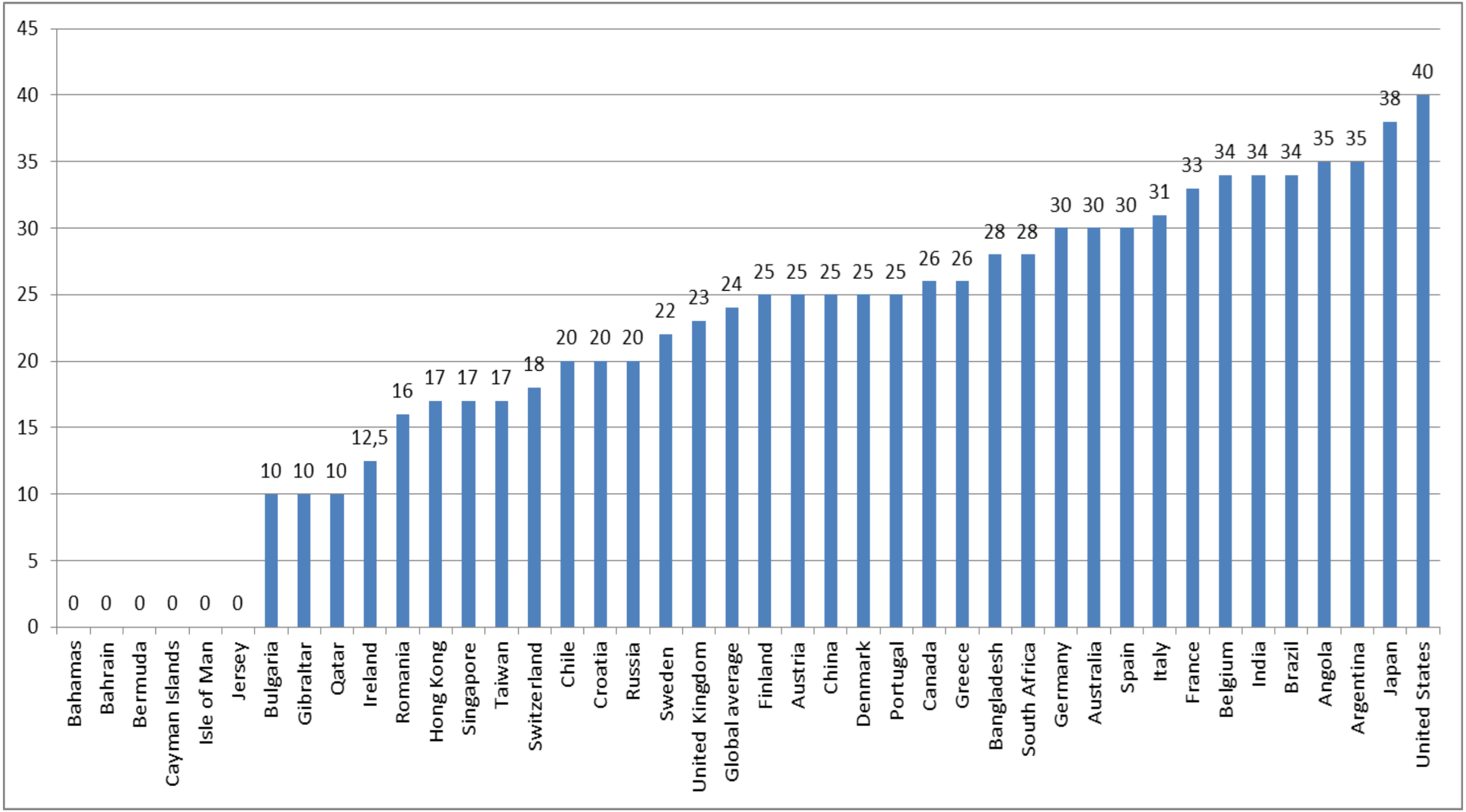
Overall income
160

Revenues	300
- Costs	-300
Operating profits	0
- Tax 60%	0
Net income	**0**

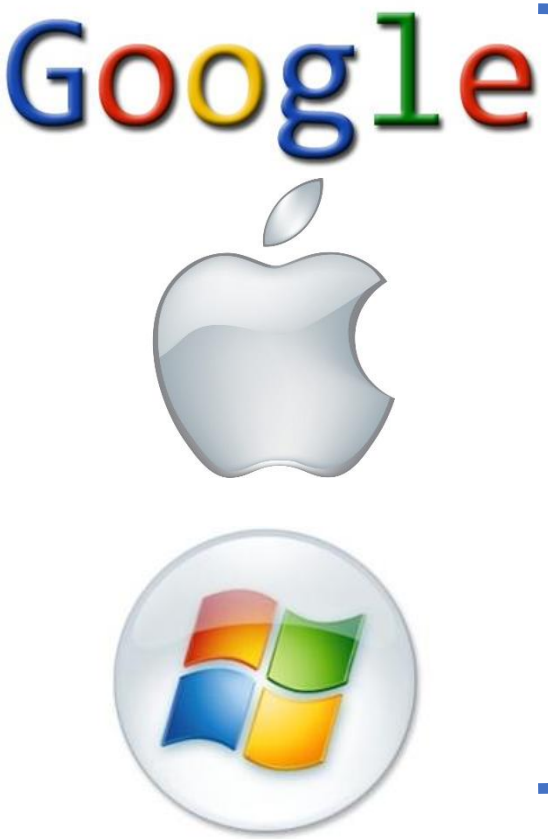
Example (Horngren chapter 23)



IRS (Internal Revenue Service = Tax authority):
claimed Transfer Price = $ 1.675 billions


45
40
35
30
25
20
15
10
5
0
0 0 0 0 0 0 10 10 10 12,5 16 17 17 17 18 20 20 20 22 23 24 25 25 25 25 25 26 26 28 28 30 30 30 31 33 34 34 34 35 35 38 40
Bahamas
Bahrain
Bermuda
Cayman Islands
Isle of Man
Jersey
Bulgaria
Gibraltar
Qatar
Ireland
Romania
Hong Kong
Singapore
Taiwan
Switzerland
Chile
Croatia
Russia
Sweden
United Kingdom
Global average
Finland
Austria
China
Denmark
Portugal
Canada
Greece
Bangladesh
South Africa
Germany
Australia
Spain
Italy
France
Belgium
India
Brazil
Angola
Argentina
Japan
United States

Which companies?



Double Irish with a
Dutch Sandwich







. . .

Transfer price regulations

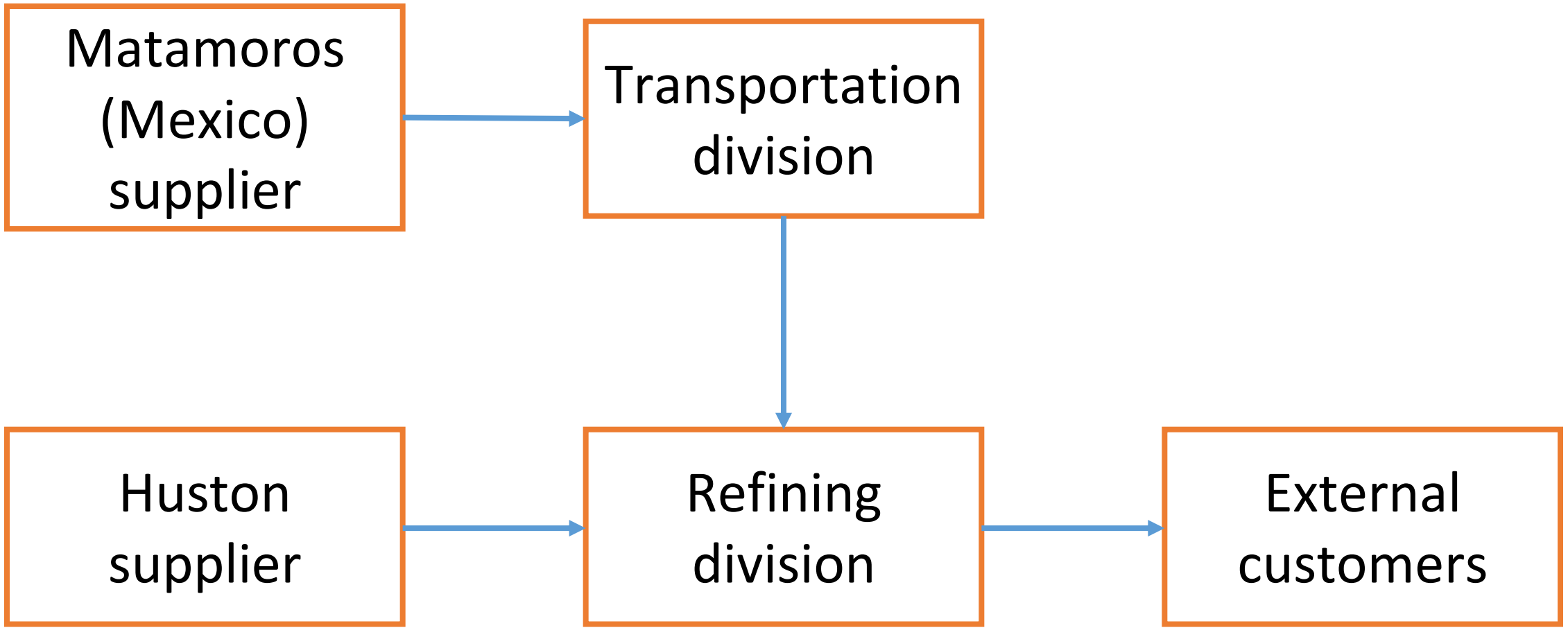
- ✓ **US IRC**: transfer price between the company and a foreign division is equal to the price that would be charged by a third party. It recognizes transfer prices can be market-based or cost-plus based (the “plus” are the margins on comparable transactions)
- ✓ **OECD Transfer Price Guidelines for multinationals and tax administrations (2017):** arm’s length principle.
- ✓ **Countries** are becoming stricter

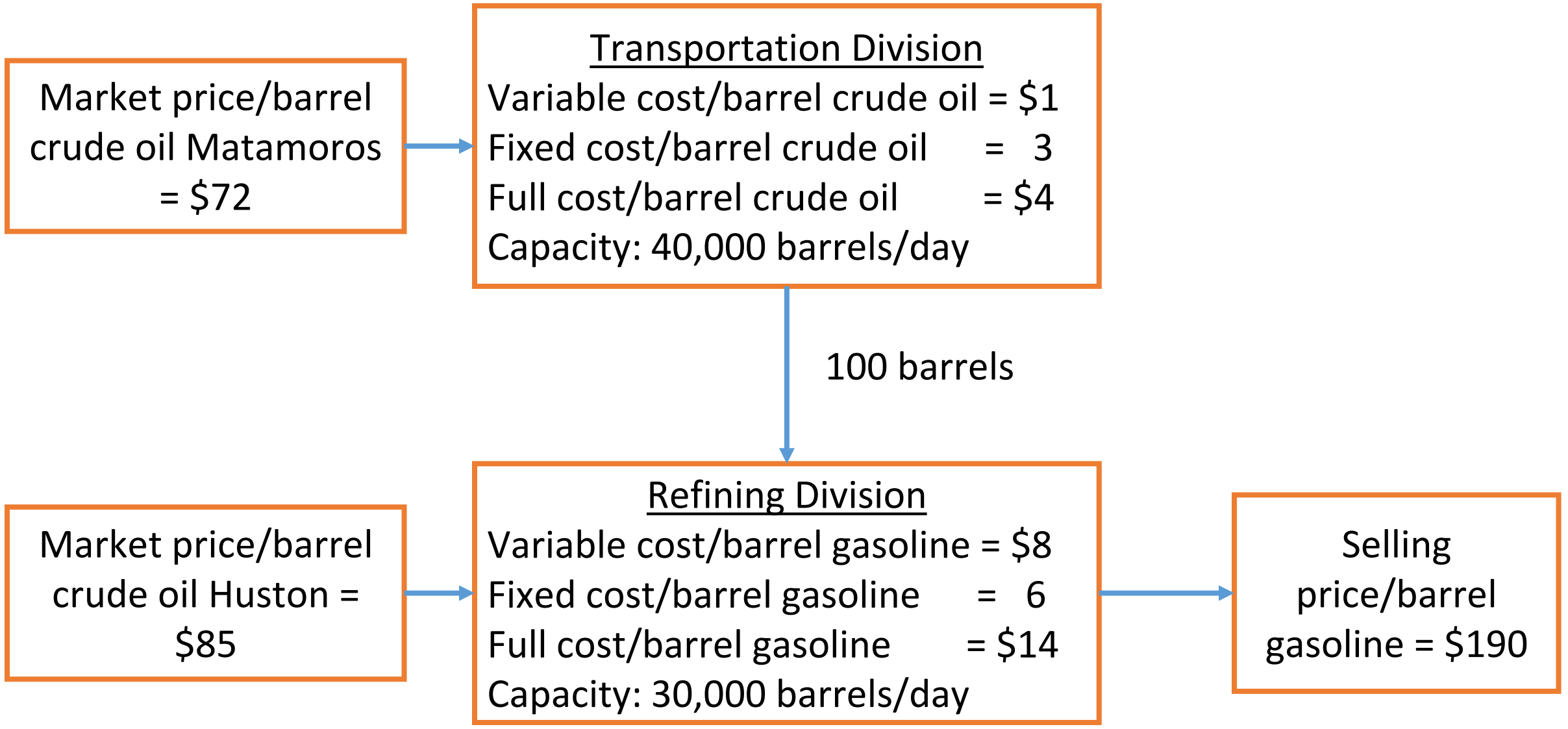
Types of transfer prices

Calculating transfer prices

- **Market-based TP**: price of product when sold to outside customers. E.g., price of a similar product publicly listed in a trade association, web site, or other.
- **Cost-based TP**: cost of producing, which can include variable production cost, variable and fixed production costs, and full costs (production costs & costs from other departments: R&D, design, marketing, distribution, customer service). Can be actual or budgeted cost. Can include a markup or profit margin.
- **Hybrid TP**: cost and market price are taken into consideration. Subunits usually negotiate the price.

Horizon Petroleum





2 barrels of oil = 1 barrel of gasoline

Horizon Petroleum (100 barrels)

Horizon Petroleum	Matamoros	Houston
Sales of Barrels crude oil / gasoline	100,00	50,00
Conversion Crude / Gasoline	2,00	1,00
Market price / Barrel crude oil	72,00	85,00
Selling price / Barrel gasoline		190,00
Transport. Division		
Variable cost / Barrel crude oil	1,00	
Fixed cost / Barrel crude oil	3,00	
Full cost / Barrel crude oil	4,00	
Capacity. Barrels / Day	40.000,00	
Refining Division		
Variable cost / Barrel gasoline	8,00	
Fixed cost / Barrel gasoline	6,00	
Full cost / Barrel gasoline	14,00	
Capacity. Barrels / Day	30.000,00	

Consolidated Operating Profit	
Horizon Petroleum	**P&L**
Sales	9.500,00
Cost of purchased oil	7.200,00
Variable costs	500,00
Fixed costs	600,00
Total Division costs	8.300,00
Company's operating profit	**1.200,00**

Horizon Petroleum

Market based TP

Mkt price crude oil	85,00	
	Transport	**Refining**
Sales	8.500,00	9.500,00
Cost of purchased oil	7.200,00	8.500,00
Variable costs	100,00	400,00
Fixed costs	300,00	300,00
Total Division costs	7.600,00	9.200,00
Division operating income	900,00	300,00
Company's operating profit	**1.200,00**	

Horizon Petroleum

Cost based TP

Transfer price 105% full cost	79,80

	Transport	Refining
Sales	7.980,00	9.500,00
Cost of purchased oil	7.200,00	7.980,00
Variable costs	100,00	400,00
Fixed costs	300,00	300,00
Total Division costs	7.600,00	8.680,00
Division operating income	380,00	820,00
Company's operating profit	**1.200,00**	

Horizon Petroleum

Hybrid based TP

A mix	82,00	
	Transport	**Refining**
Sales	8.200,00	9.500,00
Cost of purchased oil	7.200,00	8.200,00
Variable costs	100,00	400,00
Fixed costs	300,00	300,00
Total Division costs	7.600,00	8.900,00
Division operating income	600,00	600,00
Company's operating profit	**1.200,00**	

Horizon Petroleum (full capacity)

Market based TP		
Mkt price crude oil	85,00	
	Transport	**Refining**
Sales	3.400.000,00	2.850.000,00
Cost of purchased oil	2.880.000,00	2.550.000,00
Variable costs	40.000,00	120.000,00
Fixed costs	120.000,00	90.000,00
Total Division costs	3.040.000,00	2.760.000,00
Division operating income	360.000,00	90.000,00
Company's operating profit	**450.000,00**	

Cost based TP		
Transfer price 105% full cost	79,80	
	Transport	**Refining**
Internal Sales	2.394.000,00	0,00
External Sales	850.000,00	2.850.000,00
Cost of purchased oil	2.880.000,00	2.394.000,00
Variable costs	40.000,00	120.000,00
Fixed costs	120.000,00	90.000,00
Total Division costs	3.040.000,00	2.604.000,00
Division operating income	204.000,00	246.000,00
Company's operating profit	**450.000,00**	

Market-based transfer prices

Market-based transfer prices

3 conditions for using market-based transfer prices:

- Market of intermediate product is **perfectly competitive**
- **Interdependencies** between subunits are minimal
- No additional costs/benefits from selling/buying externally instead of internally

In general, companies prefer their managers to buy/sell internally

Market-based transfer prices

A perfectly competitive market exists when there is a homogeneous product with buying prices equal to selling prices and no individual buyers or sellers can affect those prices by their own actions. By using market-based transfer prices in perfectly competitive markets, a company can (1) promote goal congruence, (2) motivate management effort, (3) evaluate the performance of subunits, and (4) preserve their autonomy.

When supply outstrips demand, market prices may drop well below their historical averages. If the drop in prices is expected to be temporary, these low market prices are called "**distress prices**." Deciding whether a current market price is a distress price is often difficult.

If markets are not perfectly competitive, selling prices affect the quantity of product sold. Faced with an imperfectly competitive market, the manager of the selling division will choose a price and quantity combination for the intermediate product that maximizes the division's operating income. If the transfer price is set at this price, the buying division may find that acquiring the product is too costly and results in a loss and decide not to purchase the product.

Market-based transfer prices

✓ If transfer price < 85
✓ If transfer price > 85
✓ At 85: goal congruence

✓ Motivates unit's managers to maximize profits
✓ Serves to evaluate the viability and profitability of units

Cost-based transfer prices

Cost-based transfer prices

Helpful when:

- Markets are **not perfectly competitive**
- Market prices are unavailable
- Product is specialized
- Intermediate product not produced externally
- Higher quality of internal products/services

Full-cost transfer prices

Full-Cost & Markup Transfer Price of Transportation Division	
Cost of the barrel	72
Variable costs/barrel oil	1
Fixed costs/barrel oil	3
Full costs	76
TRANSFER PRICE	**76*1.05=79.8**

Potential drawbacks from full-cost transfer price

- If top management interferes: closer to a centralized company
- No incentive to the selling division to reduce costs
- When there is a markup, costs and profits increase by the same percentage

Hybrid transfer prices

Hybrid transfer prices

Hybrid between market and cost based prices:

- ✓Minimum transfer price = 76 (full cost without markup)
- ✓Maximum transfer price = 85 (market based transfer price)

Hybrid transfer prices

Options to reach an aceptable hybrid Price:

- ✓ Prorating de difference between max. and min. prices based on an adequate activity driver (i.e. variable costs in Horizon).
- ✓ Negotiating the Price but several factors to consider (i.e. bargaining strength os the subunits, knowledge of the market…)
- ✓ Dual pricing: uses two separate transfer-pricing methods to price each transfer from one subunit to another. Not widely used.

Negotiated pricing

- ✓ Most common hybrid method
- ✓ Cost and price information is usually the starting point in the negotiation process
- ✓ Preserves division autonomy

Class excercise

Questron Co.

The Questron company manufactures telecommunications equipment in the US. The company has marketing divisions throughout the world. A Questron marketing division located in Germany imports 100,000 broadband routers from the manufacturing division in the US. The following information is available:

US corporate tax rate	35%
German corporate tax rate	40%
German import duty	15%
Variable manufacturing cost per router ($)	275
Full manufacturing cost per router ($)	400
Selling price in Germany ($)	575

Suppose that the US and German tax authorities only allow transfer prices that are between the full manufacturing cost per unit of $400 and a market price of $475, based on comparable imports into Germany. The German import duty is charged on the price at which the product is transferred into Germany. Import duty is a deductible expense for calculating German income taxes.

1. **Calculate the net income earned by the US and German divisions from transferring 100,000 routers at a) full manufacturing cost per unit and b) at market price of comparable imports.**
2. **Which transfer price should the Questron company select to minimize its income taxes? Remember the transfer price must be between $400 and $475.**

Questron Co. (TP at manufact. Cost)

Net income (manuf. Cost)	US	Germ
Sales	40.000.000,00	57.500.000,00
Cost of goods	40.000.000,00	40.000.000,00
Import duty tax	0,00	6.000.000,00
Tax rate		*15,0%*
Gross margin	0,00	11.500.000,00
Corporate Tax	0	4.600.000,00
Tax rate	*35,0%*	*40,0%*
Net Income	**0,00**	**6.900.000,00**
	6.900.000,00	
Total Income Tax paid	*4.600.000,00*	

Questron Co. (TP at market price)

Net income (market price)	US	Germ
Sales	47.500.000,00	57.500.000,00
Cost of goods	40.000.000,00	47.500.000,00
Import duty tax	0,00	7.125.000,00
Tax rate		*15,0%*
Gross margin	7.500.000,00	2.875.000,00
Corporate Tax	2.625.000,00	1.150.000,00
Tax rate	*35,0%*	*40,0%*
Net Income	**4.875.000,00**	**1.725.000,00**
	6.600.000,00	
Total Income Tax paid	*3.775.000,00*	

Final comments

Final comments

Criteria	Market-Based	Cost-Based	Negotiated
Achieves goal congruence	Yes, when markets are competitive	Often, but not always	Yes
Motivates management effort	Yes	Yes, when based on budgeted costs; less incentive to control costs if transfers are based on actual costs	Yes
Useful for evaluating subunit performance	Yes, when markets are competitive	Difficult unless transfer price exceeds full cost and even then is somewhat arbitrary	Yes, but transfer prices are affected by bargaining strengths of the buying and selling divisions
Preserves subunit autonomy	Yes, when markets are competitive	No, because it is rule-based	Yes, because it is based on negotiations between subunits
Other factors	Market may not exist, or markets may be imperfect or in distress	Useful for determining full cost of products and services; easy to implement	Bargaining and negotiations take time and may need to be reviewed repeatedly as conditions change

Final comments

- ✓Difficult for any transfer pricing method to meet **all** advantages of TP
- ✓Survey info indicates **full-cost-based transfer pricing is most common**, then market-based then negotiated
- ✓**Main** objective of TP: Enables centers to be profit centers
- ✓Sometimes used to avoid taxes

REFERENCES

- Hongren et al : Chapter 23
- Business case, White Hills Children's Museum (assignment, 5%)
- Additional references:
  - OECD Transfer price guidelines for multinational enterprises and tax administrations. July 2017.